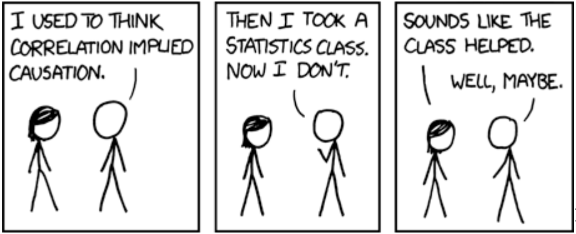
Causal Inference

STSCI / INFO / ILRST 3900

25 Aug 2026



[1]

[1] https://xkcd.com/552/

Welcome to Causal Inference!

Sam Wang

Filippo Fiocchi

Shira Mingelgrin

Why causal inference?

Why causal inference?

Reasoning about causal claims allows us to make better decisions

Why causal inference?

Most of statistics is about

- Describing the current state of the world
- Making predictions when observing a system
- "What **is** the world like?"

Why causal inference?

Most of statistics is about

- Describing the current state of the world
- Making predictions when observing a system
- "What **is** the world like?"

Correlation $\neq$ Causation

Why causal inference?

Most of statistics is about

- Describing the current state of the world
- Making predictions when observing a system
- "What **is** the world like?"

Correlation $\neq$ Causation

Causal inference provides tools to

- Understand causal mechanisms
- Make claims about what would have happened under an intervention
- Compare a hypothetical worlds where X happened to a world where everything else is the same except X did not happen
- Reason about additional assumptions
- "What **could** the world be like?"

Causality

Reasoning about interventions

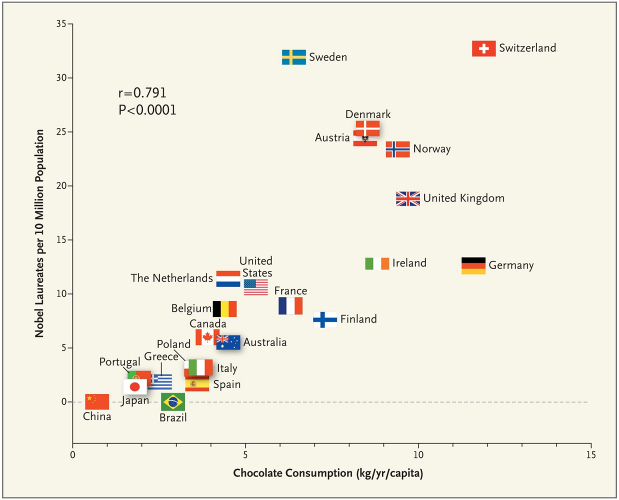


Figure: Chocolate vs Nobel Prizes from (Messerli, 2012, NEJM)

Examples of causal questions

Yam and Lopez (2019) ask if using a more aggressive 4th down strategy would **cause** NFL teams to win more games

Examples of causal questions

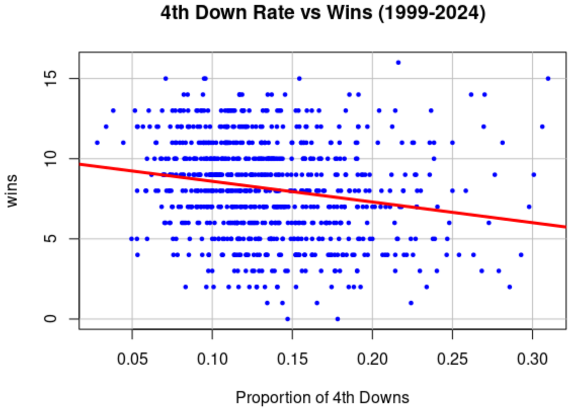
Yam and Lopez (2019) ask if using a more aggressive 4th down strategy would **cause** NFL teams to win more games

- **Observe:** Do teams which attempt more 4th down conversions end up winning more/less games?

Examples of causal questions

Yam and Lopez (2019) ask if using a more aggressive 4th down strategy would **cause** NFL teams to win more games

- **Observe:** Do teams which attempt more 4th down conversions end up winning more/less games?



Examples of causal questions

Yam and Lopez (2019) ask if using a more aggressive 4th down strategy would **cause** NFL teams to win more games

- **Observe:** Do teams which attempt more 4th down conversions end up winning more/less games?
- **Intervene:** Would a team win more games if they used a more aggressive 4th down strategy compared to a less aggressive strategy?

Examples of causal questions

Yam and Lopez (2019) ask if using a more aggressive 4th down strategy would **cause** NFL teams to win more games

- **Observe:** Do teams which attempt more 4th down conversions end up winning more/less games?
- **Intervene:** Would a team win more games if they used a more aggressive 4th down strategy compared to a less aggressive strategy?

Finding: On average, teams could add an additional .4 games by implementing a more aggressive 4th down strategy

Examples of causal questions

Kearney and Levine (2015) ask if the reality TV show "16 and Pregnant" **caused** a decrease in US teen pregnancy rates

Examples of causal questions

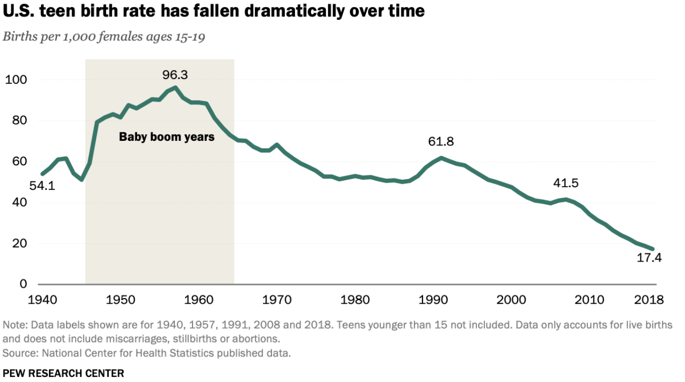
Kearney and Levine (2015) ask if the reality TV show "16 and Pregnant" **caused** a decrease in US teen pregnancy rates

- **Observe:** Did the teen pregnancy rate is decrease after the TV show aired (2009)?

Examples of causal questions

Kearney and Levine (2015) ask if the reality TV show “16 and Pregnant” **caused** a decrease in US teen pregnancy rates

- **Observe:** Did the teen pregnancy rate is decrease after the TV show aired (2009)?



Examples of causal questions

Kearney and Levine (2015) ask if the reality TV show "16 and Pregnant" **caused** a decrease in US teen pregnancy rates

- **Observe:** Did the teen pregnancy rate is decrease after the TV show aired (2009)?
- **Intervene:** Did airing the show cause the pregnancy rate to decrease more than it would have compared to if the show had not aired?

Examples of causal questions

Kearney and Levine (2015) ask if the reality TV show “16 and Pregnant” **caused** a decrease in US teen pregnancy rates

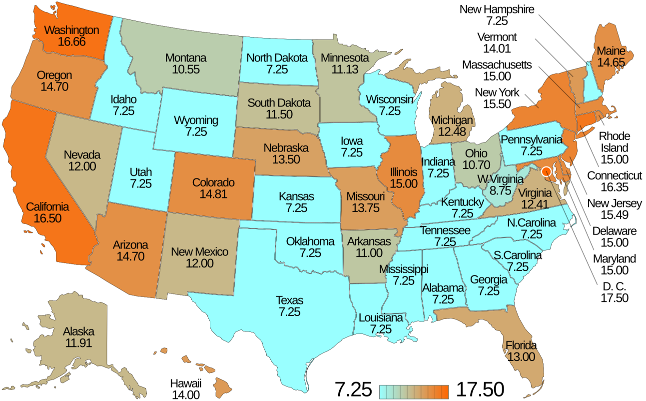
- **Observe:** Did the teen pregnancy rate is decrease after the TV show aired (2009)?
- **Intervene:** Did airing the show cause the pregnancy rate to decrease more than it would have compared to if the show had not aired?

Finding: The airing of “16 and Pregnant” caused a 4.3% decrease in teen pregnancies

Examples of causal questions

Card and Krueger (1994) ask if increasing the minimum wage **causes** in an increase or decrease to the number of employees at fast food restaurants?

State minimum wages, in dollars. Feb. 21, 2025



Source: wikimedia

Examples of causal questions

Card and Krueger (1994) ask if increasing the minimum wage **causes** in an increase or decrease to the number of employees at fast food restaurants?

- **Observe:** Is the average employment rate is higher/lower in locations with a higher minimum wage?

Examples of causal questions

Card and Krueger (1994) ask if increasing the minimum wage **causes** in an increase or decrease to the number of employees at fast food restaurants?

- **Observe:** Is the average employment rate is higher/lower in locations with a higher minimum wage?
- **Intervene:** Would increasing the minimum wage in a location cause the employment rate to be higher/lower than it would have been if minimum wage remained the same?

Examples of causal questions

Card and Krueger (1994) ask if increasing the minimum wage **causes** in an increase or decrease to the number of employees at fast food restaurants?

- **Observe:** Is the average employment rate is higher/lower in locations with a higher minimum wage?
- **Intervene:** Would increasing the minimum wage in a location cause the employment rate to be higher/lower than it would have been if minimum wage remained the same?

Examples of causal questions

Card and Krueger (1994) ask if increasing the minimum wage **causes** in an increase or decrease to the number of employees at fast food restaurants?

- **Observe:** Is the average employment rate is higher/lower in locations with a higher minimum wage?
- **Intervene:** Would increasing the minimum wage in a location cause the employment rate to be higher/lower than it would have been if minimum wage remained the same?

Finding: Increasing the minimum wage did not cause a drop in employment

Why causal inference?

Correlation $\neq$ Causation

Why causal inference?

$$\text{Correlation} \neq \text{Causation}$$

$$\text{Correlation} + \text{Assumptions} \stackrel{?}{=} \text{Causation}$$

Why causal inference?

$$\text{Correlation} \neq \text{Causation}$$

$$\text{Correlation} + \text{Assumptions} \stackrel{?}{=} \text{Causation}$$

Assumptions can be good or bad

- Assuming true things can give us more precise answers
- Assuming false things can give us misleading results
- Does the potential benefit of an assumption outweigh the potential costs?

Course objectives

As a result of participating in this course, students will be able to

- Define counterfactuals as the outcomes of hypothetical interventions
- Identify counterfactuals by causal assumptions presented in graphs
- Estimate causal quantities by pairing those assumptions with statistical evidence

What else?

- **Statistics:** What **is** the world like?
- **Causal inference:** What **could** the world be like?

What else?

- **Statistics:** What **is** the world like?
- **Causal inference:** What **could** the world be like?
- **Question:** What **should** the world be like and how do we get there?

Meet your neighbors

- Name, year, and major
- What was the highlight of your summer?
- One example of a causal question which would help you make better decisions

COURSE LOGISTICS

Who should take this course?

The course is designed for upper-division undergraduate students.

Prerequisites.
An introductory statistics course at the level of STSCI 2110, ECON 3110, or similar courses.

Familiarity with R programming language

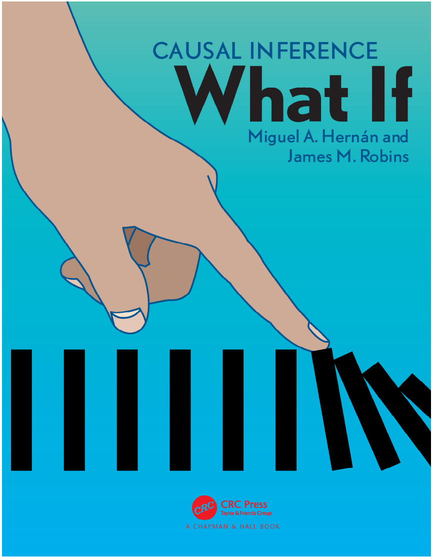
Course materials

All materials will be posted here:

causal3900.github.io

- Lectures will introduce new material
- Discussion sections will give practical details
- Post questions on Ed Discussion
- Office hours—listed at who we are page

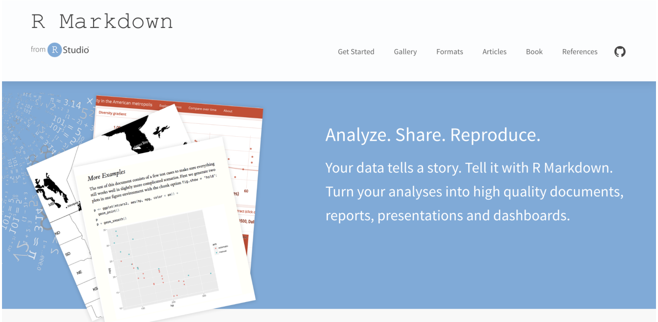
Course readings



Method of assessing student achievement

In-class assignments	10%
Class project	25%
Quizzes	65%

Typesetting



As soon as possible, you should

- Install R (statistical software)
- Install RStudio (user interface)
- Bookmark the RMarkdown cheat sheet (documentation)

Academic integrity

Each student in this course is expected to abide by the Cornell University Code of Academic Integrity. Any work submitted by a student in this course for academic credit must be the student's own work.

Generative AI tools should not be used for generating text/code on assignments unless explicitly allowed.

Students with disabilities

You belong in this course. We are happy to work with you on appropriate accomodations—see the syllabus for details about working with Student Disability Services.

Mental health and wellbeing

Your health and wellbeing are important to us!

See syllabus for links to mental health resources. We hope our course helps you thrive a Cornell, and your thriving at Cornell is far more important than anything in this course.

Questions

Card, D. and Krueger, A. B. (1994). Minimum wages and employment: A case study of the fast-food industry in new jersey and pennsylvania. *American Economic Review*, 84(4):772–793.

Kearney, M. S. and Levine, P. B. (2015). Media influences on social outcomes: The impact of mtv's 16 and pregnant on teen childbearing. *American Economic Review*, 105(12):3597–3632.

Messerli, F. H. (2012). Chocolate consumption, cognitive function, and nobel laureates. *New England Journal of Medicine*, 367(16):1562–1564.

Yam, D. R. and Lopez, M. J. (2019). What was lost? a causal estimate of fourth down behavior in the national football league. *Journal of Sports Analytics*, 5(3):153–167.